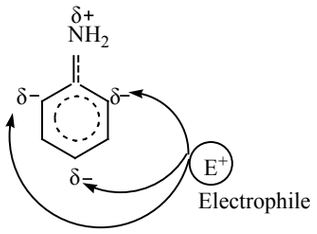
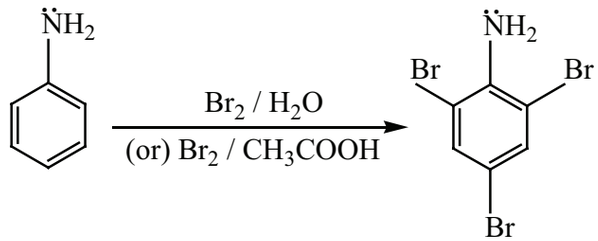
7. Electrophilic substitution reactions in Aniline

The $-\ddot{N}H_2$ group is a strong activating group. In aniline the NH_2 group is directly attached to the benzene ring, the lone pair of electrons on the nitrogen is in conjugation with benzene ring which increases the electron density at ortho and para position, thereby facilitating the electrophilic attack at ortho and para positions.



i) Bromination

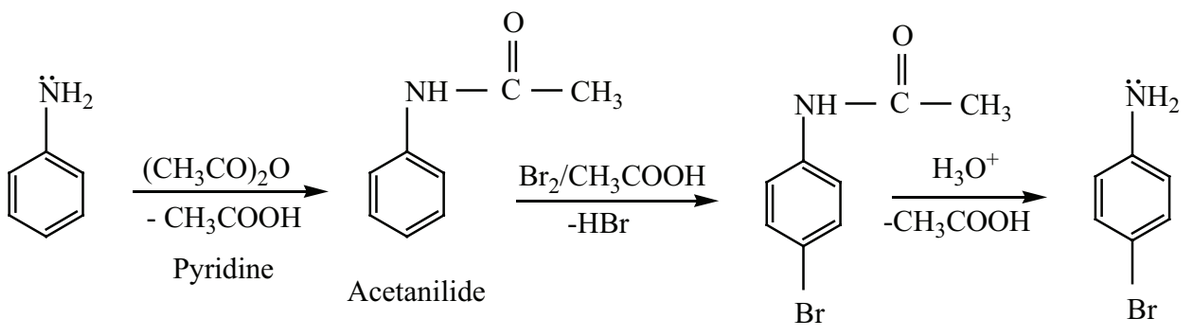
Aniline reacts with Br_2 / H_2O to give 2,4,6 – tribromo aniline a white precipitate.



Aniline

2,4,6, tribromo aniline
(white precipitate)

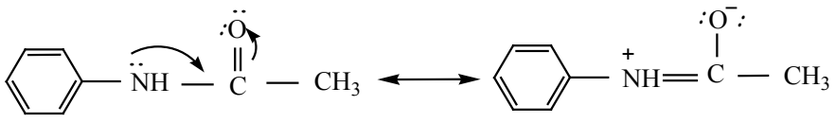
To get mono bromo compounds, - NH_2 is first acylated to reduce its activity.



p - bromo acetanilide

p-bromo aniline
(Major product)

When aniline is acylated, the lone pair of electron on nitrogen is delocalised by the neighbouring carbonyl group by resonance. Hence it is not easily available for conjugation with benzene ring.



The acetylamino group is thus less activating than the amino group in electrophilic substitution reaction.

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